

Root-Locus – Worked Problems

Practice: breakaway points, gain for a damping spec, and lead design.

Ogata, *Modern Control Engineering*, Ch. 6 (end-of-chapter style).

Step through with **Ctrl+Enter**, or render a report with **publish**.

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Problem 1: Locate the Breakaway Point

For $G(s) = \frac{K}{s(s+4)}$, find the breakaway point on the real axis and the corresponding gain.

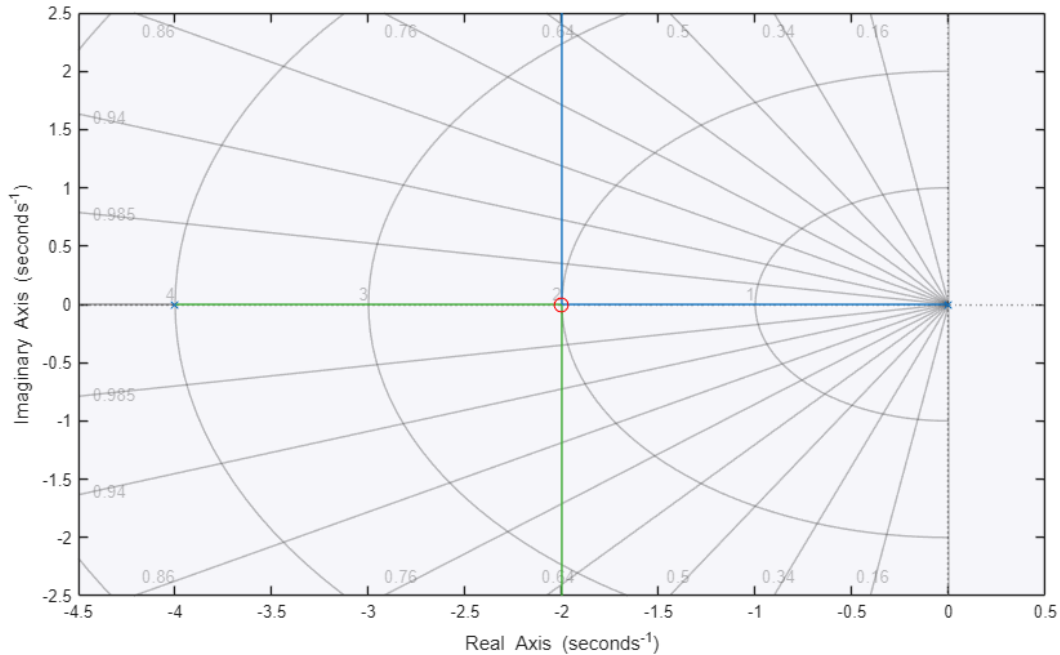
Characteristic equation: $s(s+4) + K = 0 \Rightarrow K = -s^2 - 4s$. Breakaway: $\frac{dK}{ds} = -2s - 4 = 0 \Rightarrow s = -2$. K at breakaway: $K = -(-2)^2 - 4(-2) = -4 + 8 = 4$.

```
syms s K
Ksym = -(s^2+4*s);
dKds = diff(Ksym,s);
s_breakaway = double(solve(dKds==0,s));
K_breakaway = double(subs(Ksym,s,s_breakaway));
fprintf('Breakaway point s = %.4f, K = %.4f\n', s_breakaway, K_breakaway)

G1 = tf(1,[1 4 0]);
figure
rlocus(G1)
hold on
plot(real(s_breakaway),imag(s_breakaway),'ro','MarkerSize',8)
hold off
title('Problem 1: Root Locus with Breakaway Point','Interpreter','latex','FontSize',20)
grid on
```

Breakaway point s = -2.0000, K = 4.0000

Problem 1: Root Locus with Breakaway Point



Problem 2: Gain for a Specified Damping Ratio

For the same $G(s) = \frac{K}{s(s+4)}$, find K such that the closed-loop poles have $\zeta = 0.707$.

Closed-loop char. eq.: $s^2 + 4s + K = 0$. Standard form $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \Rightarrow 2\zeta\omega_n = 4$, $\omega_n^2 = K$. With $\zeta = 0.707$: $\omega_n = 4/(2 \times 0.707) = 2.828$, so $K = \omega_n^2 \approx 8$.

```
zeta2 = 0.707;
wn2 = 4/(2*zeta2);
K2 = wn2^2;
fprintf('Required K for zeta=0.707: K = %.4f\n', K2)
```

```
T2 = feedback(K2*G1,1);
poles_2 = pole(T2)
info2 = stepinfo(T2)
```

```
Required K for zeta=0.707: K = 8.0024
```

```
poles_2 =
```

```
-2.0000 + 2.0006i
-2.0000 - 2.0006i
```

```
info2 =
```

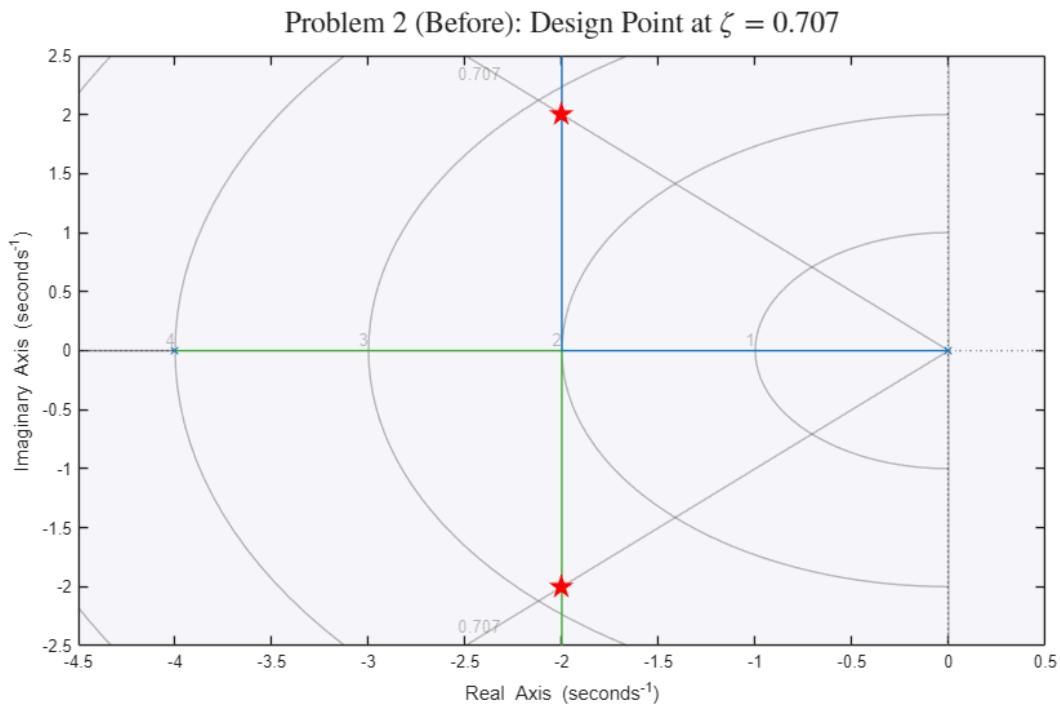
```
struct with fields:
```

```
RiseTime: 0.7595
TransientTime: 2.1079
SettlingTime: 2.1079
SettlingMin: 0.9036
SettlingMax: 1.0433
Overshoot: 4.3251
Undershoot: 0
Peak: 1.0433
PeakTime: 1.5658
```

Problem 2 – Before: Locating the Design Point on the Locus

Draw the locus together with the constant- $\zeta=0.707$ ray; their intersection is the design point, and the gain there is the K computed above. The resulting closed-loop poles are marked.

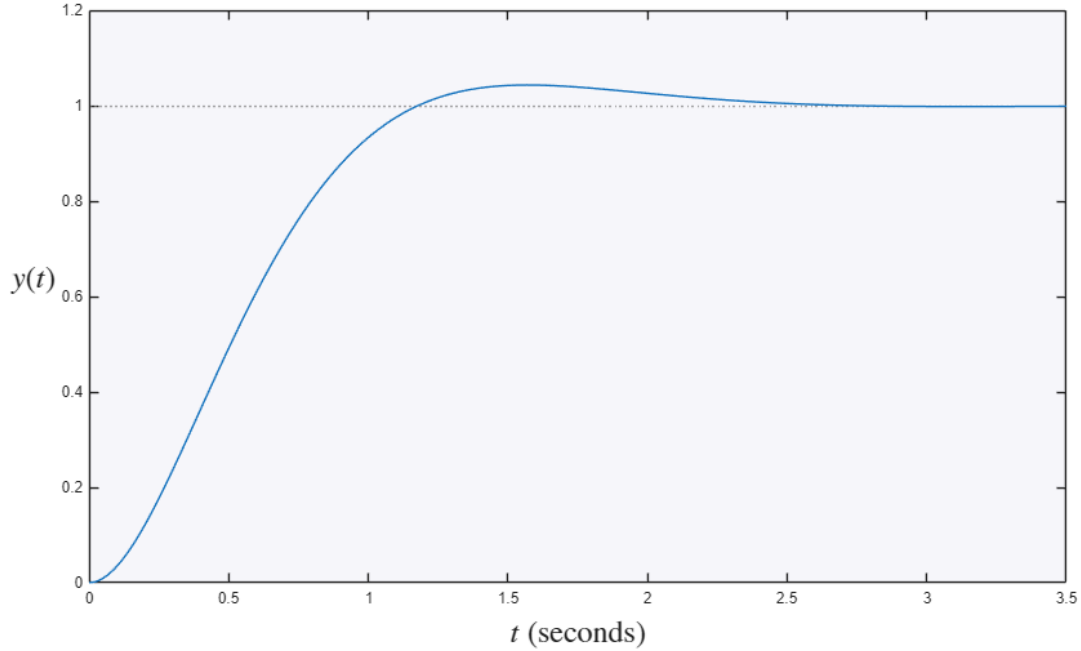
```
figure
rlocus(G1)
hold on
sgrid(0.707,[])
plot(real(poles_2),imag(poles_2),'rp','MarkerSize',13,'MarkerFaceColor','r')
hold off
title('Problem 2 (Before): Design Point at  $\zeta=0.707$ ','Interpreter','latex','FontSize',17)
grid on
```



Problem 2 – After: The Resulting Step Response

```
figure
step(T2)
title('Problem 2 (After): Step Response at the Design Point','Interpreter','latex','FontSize',18)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca,'YLabel'),'Rotation',0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```

Problem 2 (After): Step Response at the Design Point



Problem 3: Lead Compensator for a Settling-Time Spec

For $G(s) = \frac{K}{s(s+4)}$, design a lead compensator $G_c(s) = \frac{s+z_c}{s+p_c}$ so the dominant closed-loop poles achieve $\zeta = 0.5$ and $\omega_n = 6$. (Note $\omega_n = 4$ would land exactly on the uncompensated locus, requiring no lead; $\omega_n = 6$ sits to the left of it, so a genuine lead network is needed.)

```

zeta3 = 0.5; wn3 = 6;
sd3 = -zeta3*wn3 + 1j*wn3*sqrt(1-zeta3^2);

ol_poles3 = [0 -4];
phase_G3 = -sum(angle(sd3 - ol_poles3))*180/pi; % angle of G(sd3)
phi_lead3 = mod((-180 - phase_G3) + 360, 360); % lead phase to add
fprintf('Desired pole: %.4f + %.4fj\n', real(sd3), imag(sd3))
fprintf('Open-loop phase at s_d = %.2f deg, lead to supply = %.2f deg\n', ...
        phase_G3, phi_lead3)

zc3 = -real(sd3); % place zero under the desired pole's real part
angle_zero3 = angle(sd3 + zc3)*180/pi;
angle_pole3 = angle_zero3 - phi_lead3;
pc3 = -real(sd3) + imag(sd3)/tand(angle_pole3);

Gc3 = tf([1 zc3],[1 pc3]);
G3 = tf(1,[1 4 0]);
Kc3 = 1/abs(evalfr(Gc3*G3,sd3));
fprintf('Lead compensator: zero at %.4f, pole at %.4f, gain Kc = %.4f\n', -zc3, -pc3, Kc3)

G3_comp = series(Kc3*Gc3,G3);
T3_comp = feedback(G3_comp,1);

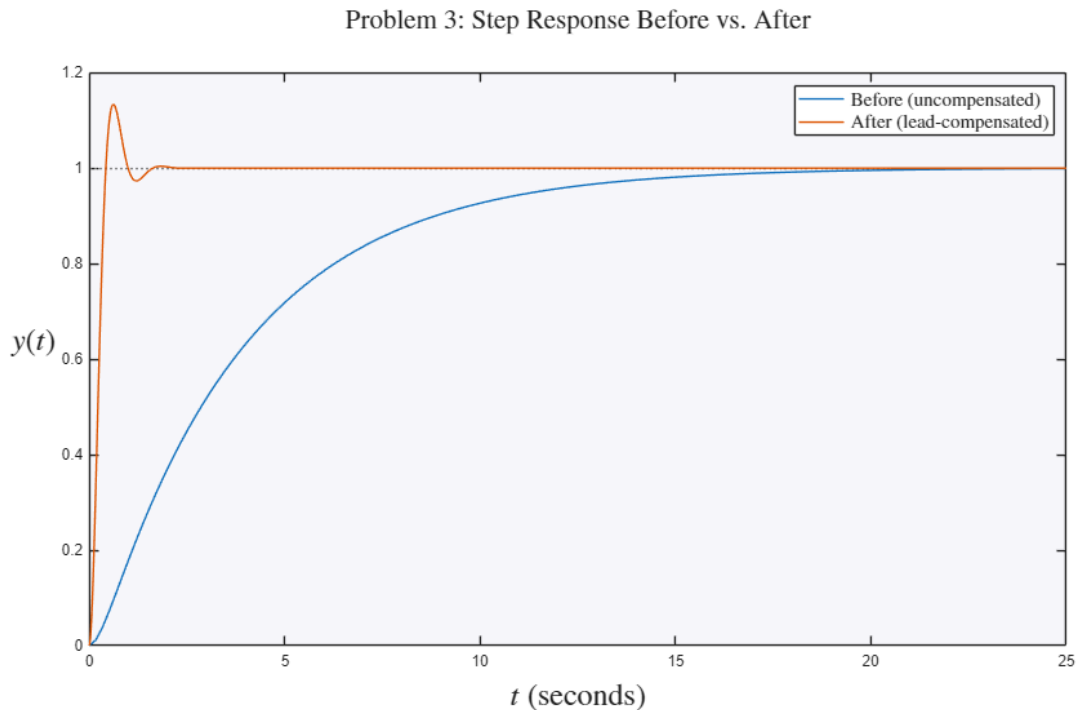
% Verify a closed-loop pole lands near the target s_d.
cl3 = pole(T3_comp);
[,i3] = min(abs(cl3 - sd3));
fprintf('Nearest closed-loop pole to s_d: %.4f + %.4fj\n', real(cl3(i3)), imag(cl3(i3)))

Desired pole: -3.0000 + 5.1962j
Open-loop phase at s_d = -199.11 deg, lead to supply = 19.11 deg
Lead compensator: zero at -3.0000, pole at -4.8000, gain Kc = 33.6000
Nearest closed-loop pole to s_d: -3.0000 + 5.1962j
    
```

Problem 3 – Before vs. After

Compare the uncompensated unit-feedback response with the lead-compensated one: the compensator speeds up and damps the response toward the $\zeta = 0.5$, $\omega_n = 6$ target.

```
T3_uncomp = feedback(G3,1);
figure
hold on
step(T3_uncomp)
step(T3_comp)
hold off
legend('Before (uncompensated)','After (lead-compensated)','Interpreter','latex','FontSize',13)
title('Problem 3: Step Response Before vs. After','Interpreter','latex','FontSize',18)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex','FontSize',20)
```



Try it yourself

- In Problem 2, target **zeta** = 0.5 instead of 0.707 and read the new required gain off the locus using the constant-zeta ray.
- In Problem 3, raise **wn3** to 8 and notice the lead network need more phase (a wider pole/zero separation).